(4000-667 cm⁻¹)
2.5 to 15 um.
Homodiatomic molecule

3.1 Introduction

Infra-red spectrum is an important record which gives sufficient information about the structure of a compound. Unlike ultraviolet spectrum which comprises of relatively few peaks, this technique provides a spectrum containing a large number of absorption bands from which a wealth of information can be derived about the structure of an organic compound. The absorption of Infra-red radiations (quantised) causes the various bands in a molecule to stretch and bend with respect to one another. The most important region for an organic chemist is 2.5 μ to 15 μ* in which molecular vibrations can be detected and measured in an infra-red spectrum and in a Raman spectrum. The ordinary infra-red region extends from 2.5 μ to 15 μ. The region ~~from 0~~.8 μ to 2.5 μ is called **Near infra-red** and that from 15 μ to 200 μ is called **Far IR Region.**

2.5 to 15 nm

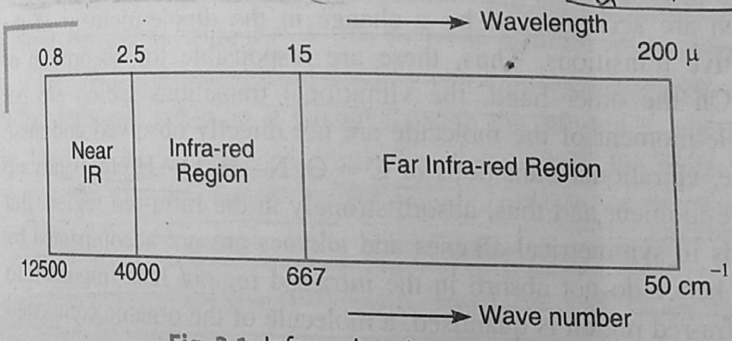
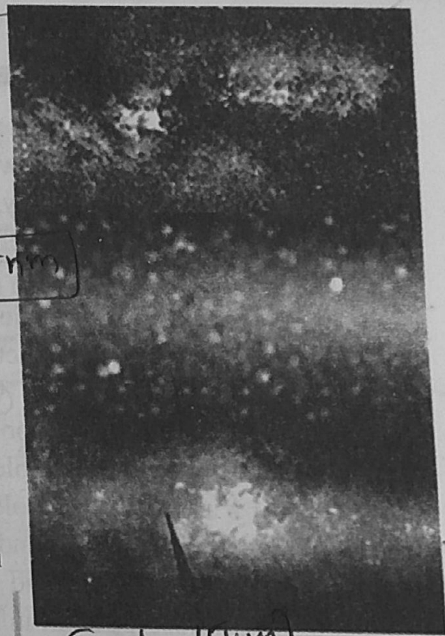


Fig. 3.1. Infra-red region.



Radiation. (2.5 to 15 um)

The absorption of Infra-red radiations can be expressed either in terms of wavelength (λ) or in wave number ($\bar{\nu}$). Mostly infra-red spectra of organic compounds are plotted as percentage transmittance against wave number. The relationship between wavelength and wave number is as follows:

0.8 nm → (2.5 to 15 nm) → UV.
400 + (um) → Nanomt

0.8

$$\text{Wave number} = \frac{1}{\text{wavelength in centimetres}}$$

If wavelength (λ) is 2.5 μ = 2.5 × 10^{-4} cm, then

$$\text{Wave number } (\bar{\nu}) = \frac{1}{2.5\times10^{-4}\,\text{cm}}$$

$$= 4000\ \text{cm}^{-1} \qquad \left[\bar{\nu} = \frac{1}{\lambda\ \text{in cm}}\right]$$

-10⁻⁷ $\frac{1}{7}$ $\frac{10}{7}$

The wavelength 15 μ corresponds to wave number equal to 667 cm^{-1}. Thus, in terms of wave number, the ordinary infra-red region covers 4000 cm^{-1} to 667 cm^{-1}.

Band intensity is either expressed in terms of absorbance (*A*) or Transmittance (*T*).

$$A = \log_{10}(1/T)$$

This technique can be employed to establish the identity of two compounds or to determine the structure of a new compound. In revealing the structure of a new compound, it is quite useful

* μ represents microns. 1 μ = 10^{-4} cm

$$\text{Wave number in cm}^{-1} = \frac{10000}{\text{wavelength in microns}}$$

to predict the presence of certain functional groups which absorb at definite frequencies. For example, the hydroxyl group in a compound absorbs at 3600-3200 cm^{-1}; carbonyl group of ketones gives a strong band at 1710 cm^{-1}. The shift in the absorption position helps in predicting the factors which cause this shift. Some of the factors which shift the absorption band for a particular group from its characteristic frequency (or wave number) are inductive effect, conjugation, angle of strain, hydrogen bonding etc. It is, thus, a very reliable technique for disclosing the identity of a compound.

vibrational fr accompaned change in dipolemome

3.2 Principle of Infra-red Spectroscopy

The absorption of Infra-red radiations causes an excitation of molecule from a lower to the higher vibrational level. We know that each vibrational level is associated with a number of closely spaced rotational levels. Clearly, the Infra-red spectra is considered as **vibrational-rotational spectra.** All the bonds in a molecule are not capable of absorbing infra-red energy but only those bonds which are accompanied by **a change** in **dipole moment** will absorb in the infra-red region. Such vibrational transitions which are accompanied by a change in the dipole-moment of the molecule are called infra-red active transitions. Thus, these are responsible for absorption of energy in the Infra-red region. On the other hand, the vibrational transitions which are not accompanied by a change in dipole-moment of the molecule are not directly observed and these are Infra-red inactive. For example, vibrational transitions of C = O, N—H, O—H etc. bands are accompanied by a change in dipole-moment and thus, absorb strongly in the Infra-red region. But transitions in Carbon-Carbon bonds in symmetrical alkenes and alkynes are not accompanied by the change in dipole-moment and hence do not absorb in the infra-red region. It is important to note that since the absorption in infra-red region is quantised, a molecule of the organic compound will show a number of peaks in the infra-red region.

3.3 Theory—Molecular Vibrations

Absorption in the infra-red region is due to the changes in the vibrational and rotational levels (see Fig. 3.2). When radiations with frequency range less than 100 cm^{-1} are absorbed, molecular rotation takes place in the substance. As this absorption is quantised, discrete lines are formed in the spectrum due to molecular rotation. Molecular vibrations are set in, when more energetic radiation in the region 10^4 to 10^2 cm^{-1} are passed through the sample of the substance. The absorption causing molecular vibration is also quantised. Clearly, a single vibrational energy change is accompanied by a large number of rotational energy changes. Thus, the vibrational spectra appear as vibrational-rotational bands. In the Infra-red spectroscopy (Region 2.5 μ – 15 μ), the absorbed energy brings about predominant changes in the vibrational energy which depends upon:

(*i*) Masses of the atoms present in a molecule,

(*ii*) Strength of the bonds, and

(*iii*) The arrangement of atoms within the molecule.

It has been found that no two compounds except the enantiomers can have similar Infra-red spectra.

It may be noted that the atoms in a molecule are not held rigidly. The molecule may be visualised as consisting of balls of different sizes tied with springs of varying strengths. Here balls and springs correspond to atoms and chemical bonds respectively. When Infra-red light is passed through the sample, the vibrational and the rotational energies of the molecules are increased. Two kinds of fundamental vibrations are:

(*a*) **Stretching.** In this type of vibrations, the distance between the two atoms increases or decreases but the atoms remain in the same bond axis.

stretching - (high)

(3600–3200)

2.5 to 15

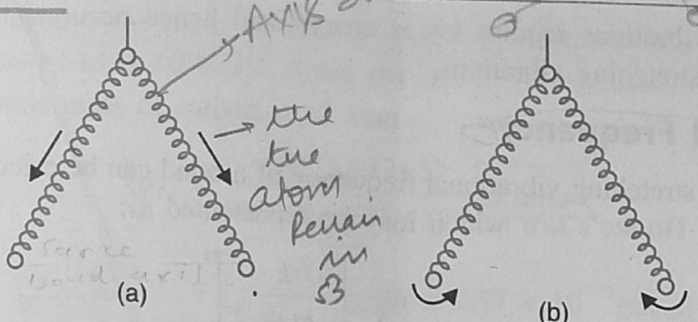


Fig. 3.2. Types of vibrations : (*a*) Stretching, (*b*) Bending.

(*b*) **Bending.** In this type of vibrations, the positions of the atoms change with respect to the original bond axis. We know that more energy is required to stretch a spring than that required to bend it. Thus, we can safely say that stretching absorptions of a bond appear at high frequencies (higher energy) as compared to the bending absorptions of the same bond.

The various stretching and bending vibrations of a bond occur at certain quantised frequencies. When Infra-red radiation is passed through the substance, energy is absorbed and the amplitude of that vibration is increased. From the excited state, the molecule returns to the ground state by the release of extra energy by rotational, collision or translational processes. As a result, the temperature of the sample under investigation increases.

Types of stretching vibrations. There are two types of stretching vibrations:

(*i*) **Symmetric stretching.** In this type, the movement of the atoms with respect to a particular atom in a molecule is in the same direction.

(*ii*) **Asymmetric stretching.** In these vibrations, one atom approaches the central atom while the other departs from it.

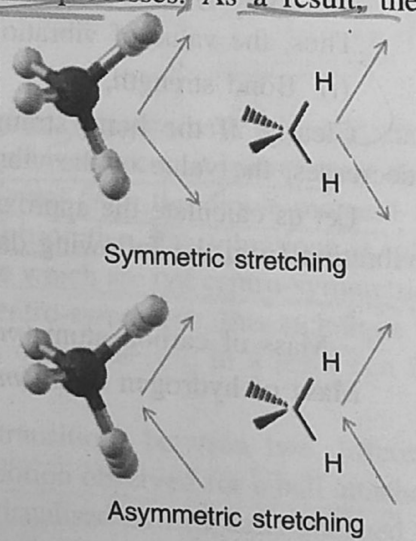


Fig. 3.3. Types of stretching vibrations.

Types of bending vibrations. (*Deformations*). Bending vibrations are of four types:

(*i*) **Scissoring.** In this type, two-atoms approach each other.

(*ii*) **Rocking.** In this type, the movement of the atoms takes place in the same direction.

(*iii*) **Wagging.** Two atoms move 'up and down' the plane with respect to the central atom.

(*iv*) **Twisting.** In this type, one of the atoms moves up the plane while the other moves down the plane with respect to the central atoms.

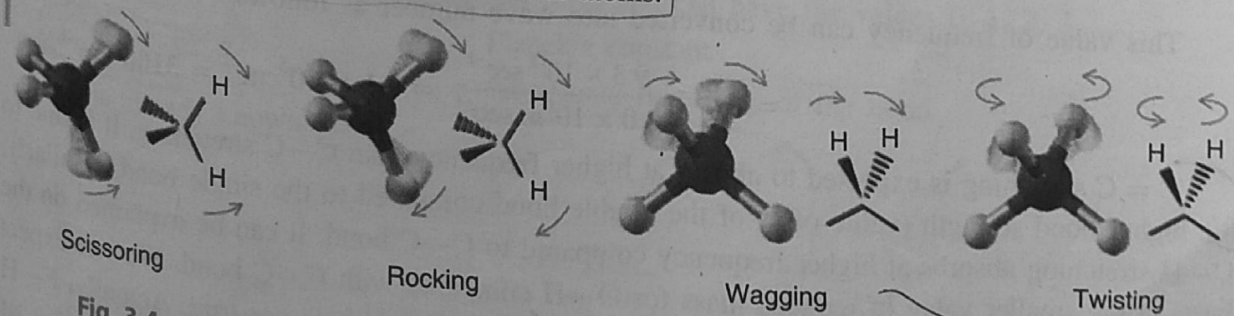


Fig. 3.4. Types of bending vibrations. (*a*) Scissoring, (*b*) Rocking, (*c*) Wagging, (*d*) Twisting

Note. Bending vibrations require lesser energy and hence occur at higher wavelengths or lower numbers than stretching vibrations.

3.4 Vibrational Frequency

The value of the stretching vibrational frequency of a bond can be calculated fairly accurately by the application of **Hooke's law** which may be represented as:

$$\frac{v}{c} = \bar{v} = \frac{1}{2\pi c}\left[\frac{k}{\frac{m_1 m_2}{m_1+m_2}}\right]^{1/2}$$

$$= \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}$$

$\frac{v}{c} = \bar{v} = \frac{1}{2\pi c}\left[\frac{k}{\frac{m_1 m_2}{m_1+m_2}}\right] = \frac{k}{2\pi\mu c}$

where μ is the reduced mass.

m_1 and m_2 are the masses of the atoms concerned in grams in a particular bond.

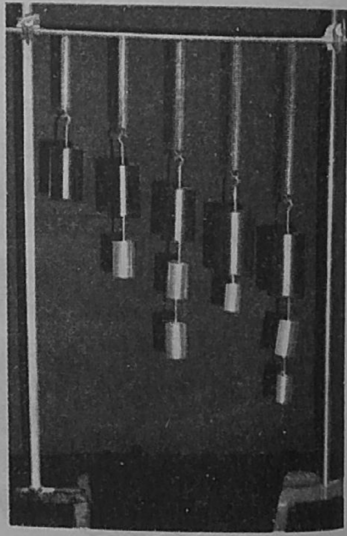
k = Force constant of the bond and relates to the strength of the bond. For a single bond, it is approximately 5×10^5 gm sec^{-2}. It becomes double and triple for the double and triple bonds respectively.

c = Velocity of the radiation = 2.998×10^{10} cm sec^{-1}

Thus, the value of vibrational frequency or wave number depends upon:

(*i*) Bond strength, and (*ii*) Reduced mass.

Clearly, if the bond strength increases or the reduced mass decreases, the value of the vibrational frequency increases.



Hooke's Law

Let us calculate the approximate frequency of C—H stretching vibration from the following data:

$$k = 5 \times 10^5 \text{ gm sec}^{-2}$$

Mass of carbon atom (m_1) = 20×10^{-24} gm

Mass of hydrogen atom (m_2) = 1.6×10^{-24} gm

$$v = \frac{1}{2\pi}\sqrt{\frac{k}{\frac{m_1 m_2}{m_1+m_2}}}$$

$$= \frac{7}{2\times 22}\left(\frac{5\times 10^5 \text{ gm sec}^{-2}}{\frac{20\times 10^{-24}\text{ gm} \times 1.6\times 10^{-24}\text{ gm}}{(20+1.6)10^{-24}\text{ gm}}}\right)^{1/2}$$

$$= 9.3 \times 10^{13} \text{ sec}^{-1}$$

This value of frequency can be converted into wave number as follows:

$$\bar{v} = \frac{v}{c} = \frac{9.3\times 10^{13}\text{ sec}^{-1}}{3.0\times 10^8 \text{ m sec}^{-1}} = 3.1\times 10^5 \text{ m}^{-1} = \mathbf{3100\ cm^{-1}}$$

C = C stretching is expected to absorb at higher frequency than C—C stretching. It is due to the higher bond strength (value of k) of the double bond compared to the single bond. Similarly, O—H stretching absorbs at higher frequency compared to C—C bond. It can be explained on the basis of the smaller value of reduced mass for O—H compared with C—C bond. We can expect O—H to absorb at higher frequency as compared to F—H. But this is not true. Actually, F—H absorbs at the higher frequency. This can be explained due to the higher electronegativity of fluorine compared to that of oxygen.

EXAMPLE 1. *Calculate the wave number of stretching vibration of a carbon-carbon double bond. Give force constant ($k = 10 \times 10^5$ dynes cm^{-1})*

SOLUTION. Atomic mass of carbon = 12 amu

12×

Reduced mass $(\mu) = \dfrac{m_1 m_2}{m_1 + m_2} = \dfrac{12 \times 12}{12+12} = \boxed{6\ amu}$

$$= \boxed{\frac{6}{6.02 \times 10^{23}}} \text{ gm} = 9.97 \times 10^{-24} \text{ gm}$$

$\dfrac{K}{2\pi mc}$

$\boxed{\dfrac{1}{2\pi c}\sqrt{\dfrac{K}{m}}}$

Wave number $\bar{\nu} = \dfrac{1}{2\pi c}\sqrt{\dfrac{k}{\mu}}$

higher wave

$$\bar{\nu} = \frac{1}{2 \times 3.142 \times 3 \times 10^{10} \text{ cm s}^{-1}} \sqrt{\frac{10 \times 10^5 \text{ gm sec}^{-2}}{\dfrac{6 \text{ gm}}{6.02 \times 10^{23}}}}$$

Bending vibration require lower energy

$$= \frac{1}{18.852 \times 10^{10} \text{ cm s}^{-1}} \times \sqrt{10.03 \times 10^{28}} = 0.1680 \times 10^4 \text{ cm}^{-1}$$

$$\boxed{= \mathbf{1680\ cm^{-1}}.}$$

(Interact with)

Oscillating dipole moment

oscillating electric vector of infrared

3.5 Number of Fundamental Vibrations

The Infra-red light is absorbed when the oscillating dipole-moment interacts with the oscillating electric vector of an Infra-red beam. For this interaction or absorption to occur, it is important that the dipole-moment at one extreme of the vibration must be different from the dipole-moment at the other extreme of the vibration in a molecule. For Infra-red absorption, the vibrations should not be centro-symmetric. Only those vibrations are Infra-red active which are not centro-symmetric. As most of the functional groups in organic chemistry are not centro-symmetric, this technique is most informative to organic chemists. The symmetry properties of a molecule in a solid can be different from those of an isolated molecule.

The Infra-red spectrum of a molecule results due to the transitions between two different vibrational energy levels. The vibrational motion resembles the motion observed for a ball attached to a spring, i.e., harmonic oscillator. A chemical bond can be visualised as two balls attached to a spring. But it differs from this system since in Infra-red certain vibrational energy levels only are allowed in molecules. The vibrational energy of a chemical bond is quantised and can have the value.

$$\boxed{E_{vib} = \left(V + \frac{1}{2}\right) h\nu}$$

Non-n Centro Symetric vibration

where V is the number of the vibrational level and can have the values 0, 1, 2, 3........

h = Planck's constant

and ν = vibrational frequency of the bond.

It is already known that

$$\nu = \frac{1}{2\pi}\sqrt{\frac{k}{\dfrac{m_1 m_2}{m_1 + m_2}}}$$

The energy difference between the two vibrational energy levels can be written as

$$\Delta E_{vib} = h\nu$$

At ordinary temperature, where molecules are in their lowest vibrational energy levels, the potential energy diagram approximates that of a harmonic oscillator. But at higher temperatures, the deviations do occur (see Fig. 3.5).

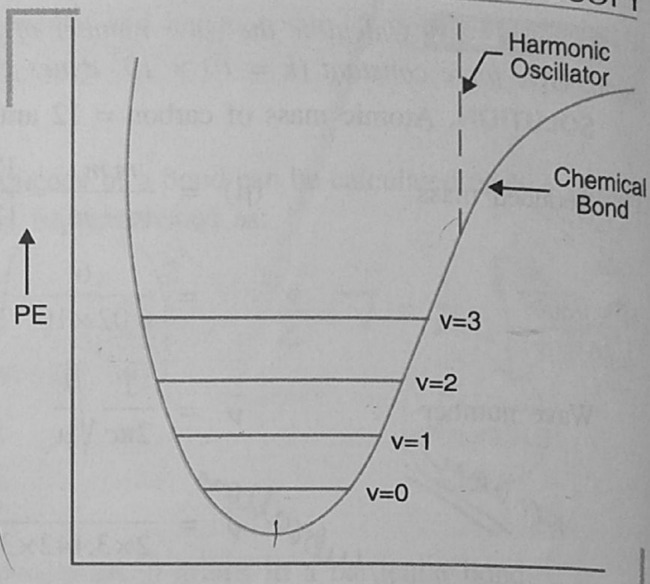


Fig. 3.5. Potential Energy diagram.

Absorption of radiation with energy equal to the difference between two vibrational energy levels (ΔE_{vib}) will cause a vibrational transition to occur. Radiation with energy sufficient to cause vibrational transition is found in the Infra-red region of the electromagnetic spectrum.

Transitions from the ground state ($V = 0$) to the first excited state ($V = 1$) absorb light strongly and give rise to intense bands called the Fundamental bands. The energy difference (ΔE_{vib}) between the lowest possible energy level of a bond and the next higher energy level is given as:

$$\Delta E_{vib} = E_{vib(V=1)} - E_{vib(V=0)}$$

$$= \left(1+\frac{1}{2}\right)h\nu - \left(0+\frac{1}{2}\right)h\nu$$

$$= \frac{3}{2}h\nu - \frac{1}{2}h\nu = h\nu$$

ΔEvib – Fund(v=0) (1+½)hν – (0+½) (hν)

This gives the frequency of a fundamental band. Transitions from the ground state ($V = 0$) to the second excited state ($V = 2$) with the absorption of Infra-red radiation give rise to weak bands, called overtones.

Assuming that all the vibrational bands are equally spaced (in fact, these are not), the energy of the first overtone is given by

$$\Delta E_{vib} = E_{vib(V=2)} - E_{vib(V=0)}$$

$$= \left(2+\frac{1}{2}\right)h\nu - \left(0+\frac{1}{2}\right)h\nu$$

$$= 2\ h\nu$$

Polyatomic molecules may exhibit more than one fundamental vibrational absorption bands. The number of these fundamental bands is related to the **degrees of freedom** in a molecule. The number of degrees of freedom is equal to the sum of the co-ordinates necessary to locate all the atoms of a molecule in space. Each atom has three degrees of freedom corresponding to the three cartesian co-ordinates (X, Y, Z) necessary to describe its position relative to other atoms in a molecule. An isolated atom which is considered as a point mass has only translational degrees of freedom. It cannot have vibrational and rotational degrees of freedom. When atoms combine to form a molecule, no degrees of freedom are lost, *i.e.*, the total number of degrees of freedom of a molecule will be equal to $3n$ where n is the number of atoms in a molecule. A molecule which is of finite dimensions will, thus, be made up of rotational, vibrational and translational degrees of freedom. So,

$3n$ degrees of freedom = Translational + Rotational + Vibrational.

Rotational degrees of freedom result from the rotation of a molecule about an axis through the centre of gravity. Since we are concerned only with the number of fundamental vibrational modes of a molecule, so we calculate only the number of vibrational degrees of freedom of a molecule. Since only three co-ordinates are necessary to locate a molecule in space, we say that a **molecule has always three translational degrees of freedom.**

In case of linear molecule, there are only two degrees of rotation. It is due to the fact that the rotation of such a molecule about its axis of linearity does not bring about any change in the position of the atoms while rotation about the other two axes changes the position of the atoms. Thus, for a linear molecule of n atoms.

Total degrees of freedom	$= 3n$
Translational degrees of freedom	$= 3$
Rotational degrees of freedom	$= 2$
$\therefore$ Vibrational degrees of freedom	$= 3n - 3 - 2 = \mathbf{3n - 5}$

linear molecule

Total degree of freedom

$(3n-5)$

Each vibrational degree of freedom corresponds to the fundamental mode of vibration and each fundamental mode corresponds to a band. Hence, theoretically there will be $3n - 5$ possible fundamental bands for the linear molecules.

In a linear molecule of carbon dioxide (CO_2), the number of vibrational degrees of freedom can be calculated as follows:

Number of atoms (n)	$= 3$
Total degrees of freedom	$= 3n = 3 \times 3 = 9$
Translational	$= 3$
Rotational	$= 2$
$\therefore$ Vibrational degrees of freedom	$= 9 - 3 - 2 = 4$

til $(3n-5)$

$[3n-5]$

Hence, for carbon dioxide molecule, the theoretical number of fundamental bands should be equal to four.

In the case of non-linear molecule, there are three degrees of rotation as the rotation about all the three axes (X, Y, Z) will result in a change in the position of the atoms. So, for a non-linear molecule of n atoms, the vibrational degrees of freedom can be calculated as follows:

Total degrees of freedom	$= 3n$
Translational degrees of freedom	$= 3$
Rotational degrees of freedom	$= 3$
$\therefore$ Vibrational degrees of freedom	$= 3n - 3 - 3 = \mathbf{3n - 6}$

$(3n-6)$

Thus, in a non-linear molecule, C_6H_6, the number of vibrational degrees of freedom can be calculated as follows:

Number of atoms (n)	$= 12$
Total degrees of freedom	$= 3 \times 12 = 36$
Translational	$= 3$
Rotational	$= 3$
$\therefore$ Vibrational degrees of freedom	$= 36 - 3 - 3 = \mathbf{30}$

fundamental vibration too weak to be too weak to be observed as Band

So, theoretically there should be 30 fundamental bands in the Infra-red spectrum of benzene.

It has been observed that the theoretical number of fundamental vibrations are seldom obtained. It is because of the following reasons:

(*i*) Fundamental vibrations that fall outside the region under investigation, *i.e.*, 2.5 to 15 μ.

(*ii*) Fundamental vibrations that are too weak to be observed as bands. →

(*iii*) Fundamental vibrations that are so close that they overlap, *i.e.*, degenerate vibrations.

(*iv*) Certain vibrational bands do not appear for want of the required change in the dipole-moment in a molecule.

dy

(*v*) There may appear some additional bands called combination bands, difference bands and overtones. Thus, due to these, a large number of bands will be observed as compared to the theoretical number. If there are two fundamental bands at x and y, then the additional bands can be expected at:

(*i*) $2x$, $2y$ (Overtones) 2x, 2y x+y

(*ii*) $x + y$, $x + 2y$, $2x + y$ etc. (Combination bands) Overtone and Combine

(*iii*) $(x - y)$, $(x - 2y)$, $(2y - x)$ etc. (Difference bands)

It may be noted that these additional bands are usually 10-100 times less intense as compared to the fundamental bands.

Note. Absorptions due to bending vibrations occur at frequencies below 1500 cm^{-1}. These are not very useful for structural information as confusion of one with the other may occur. Moreover, bands are obscured due to overtones and combination bands. The region below 1500 cm^{-1} is called **Finger print region.** This region is important for comparing the identity of the two compounds and also for the detection of certain functional groups like esters, ethers, type of disubstitution on the benzene ring etc.

Bending (1500cm⁻¹) below

3.6 Selection Rules (Active and Forbidden Vibrations)

Infra-red light is absorbed only when a change in dipole character of the molecule takes place. Complete symmetry about a bond eliminates some absorption bands. Clearly, some of the fundamental vibrations are Infra-red active and some are not. It is governed by the selection rules which are explained below:

(*i*) If a molecule has a centre of symmetry, then the vibrations are centrosymmetric and are **inactive** in the Infra-red but are active in the Raman.

(*ii*) The vibrations which are not centrosymmetric are active in Infra-red but inactive in Raman.

Since in most of the organic compounds, the functional groups are not centrosymmetric. Infra-red spectroscopy is most informative. Consider various vibrations in case of carbon dioxide.

$\overleftarrow{O} = C = \overrightarrow{O}$

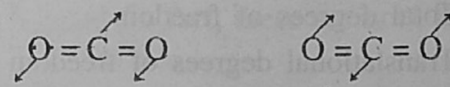
(*i*)

Symm Stretching
DM = 0. IR inactive

$\overrightarrow{O} = C = \overrightarrow{O}$

(*ii*)

Asymm Stretching
$\nu = 2350\ cm^{-1}$
IR inactive



(*iii*) (*iv*)

Bending vibrations
$\nu = 667\ cm^{-1}$

The above vibrations are all fundamental vibrations of carbon dioxide. Since (*i*) does not give rise to any change in dipole-moment, it is infra-red inactive. Asymmetric stretching causes, a net change in dipole-moment and thus is infra-red active and absorbs at 2350 cm^{-1} vibrations (*iii*) and (*iv*) are said to be degenerate. The bending of bonds in the molecule are identical but occur in perpendicular planes and thus appear in the same position ($\nu = 667\ cm^{-1}$) in the spectrum. Thus the spectrum of carbon dioxide consists of two bands (*i*) 2350 cm^{-1} due to asymmetric stretching and (*ii*) 667 cm^{-1} due to bending vibrations.

Intensity and position of Infra-red absorption bands. The intensity of a fundamental absorption depends upon the difference between the dipole moments of the molecule in the ground state and the vibrational excited state. Greater the difference, more is the intensity of absorption. In case, there is no change in the dipole moment, then the vibrational mode is said to be Infra-red inactive. Since the intensity of absorption band in infra-red spectroscopy cannot be measured with greater accuracy, it is sufficient to know whether the absorption is strong, medium or weak. If the most intense peak is assigned an intensity of 100%, then the relative intensities of other peaks can be estimated. Frequency of absorption is generally expressed in terms of wave number (cm^{-1}) as wave number is directly proportional to frequency as well as energy.

3.7 Factors Influencing Vibrational Frequencies

We know that the probable frequency or wave-number of absorption can be calculated by the application of Hooke's law. It has been found that the calculated value of frequency of absorption for a particular bond is never exactly equal to its experimental value. The difference arises from the fact that vibration of each group is influenced by the structure of the molecule in the immediate neighbourhood of the bond. If a particular vibration is absolutely free from any influence, then the infra-red spectrum would tell us whether a certain group is/or is not present in the molecule. Moreover, the value of absorption frequency is also shifted since the force constant of a bond changes with its electronic structure. There are many factors which are responsible for vibrational shifts and one factor cannot be isolated from another. Frequency shifts also take place on working with the same substance in different states (solids, liquids or vapours). In the vapour state, a substance, usually absorbs at higher frequency as compared to that when it is in the liquid and solid states. Following are some of the factors responsible for shifting the vibrational frequencies from their normal values:

1. **Coupled Vibrations and Fermi Resonance.** We expect one stretching absorption frequency for an isolated C—H bond but in the case of methylene (—CH_2—) group, two absorptions occur which correspond to symmetric and asymmetric vibrations as follows:

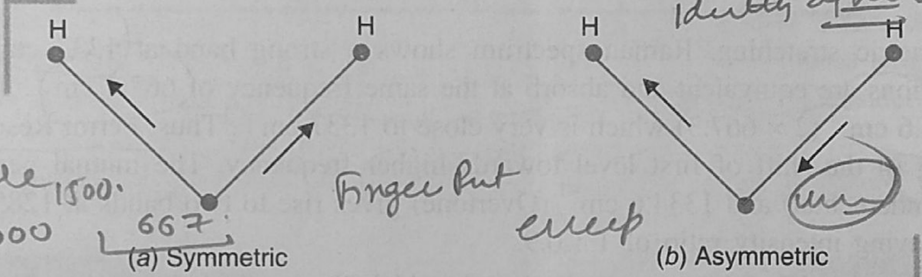


Fig. 3.6. Stretching vibrations in methylene group.

In such cases, asymmetric vibrations always occur at higher wave number compared with the symmetric vibrations. These are called coupled vibrations since these vibrations occur at different frequencies than that required for an isolated C—H stretching. Similarly, coupled vibrations of —CH_3 group take place at different frequencies compared to —CH_2— group. For methyl group, symmetric vibrations are as follows:

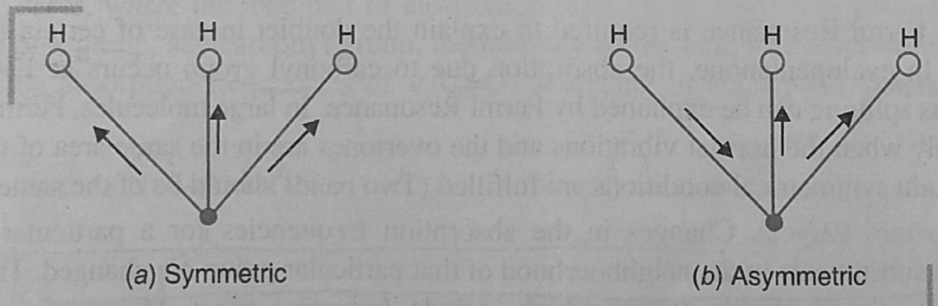


Fig. 3.7. Types of stretching vibrations in methyl group.

Sometimes, it happens that two different vibrational levels have nearly the same energy. If such vibrations belong to the same species (as in the case of —CH_2— group or —CH_3 group), then a mutual perturbation of energy may occur, resulting in the shift of one towards lower frequency and the other towards higher frequency. It is also accompanied by a substantial increase in the intensity of the respective bands.

Acid anhydrides show two C=O stretching absorptions between 1850-1800 and 1790-1745 cm^{-1} with a difference of about 65 cm^{-1}. This can be explained due to the symmetric and asymmetric stretching. In Infra-red spectrum, absorption bands are spread over a wide range of frequencies. It may happen that the energy of an overtone level chances to coincide with the fundamental mode of different vibration. A type of resonance occurs as in the case of coupled pendulums. This type of resonance is called **Fermi Resonance.** This can be explained by saying that a molecule transfers its energy from fundamental to overtone and back again. Quantum mechanically, resonance pushes the two levels apart and mixes their character so that each level becomes partly fundamental and partly overtone in character. Thus, this type of resonance gives rise to a pair of transitions of equal intensity.

This phenomenon was first observed by Enrico Fermi in the case of carbon dioxide. Carbon dioxide molecule (Triatomic) is linear and four fundamental vibrations are expected for it. Out of these, symmetric stretching vibration is Infra-red inactive since it produces no change in the dipole-moment (**D.M.**) of the molecule.

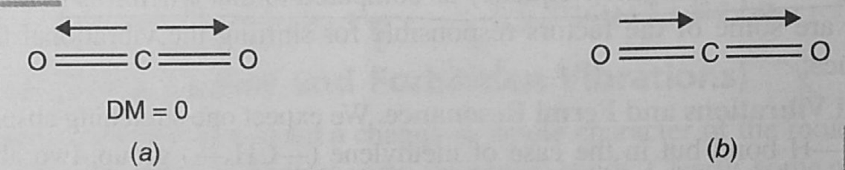


Fig. 3.8. (*a*) Symmetric stretching, (*b*) Asymmetric stretching.

For symmetric stretching, Raman spectrum shows a strong band at 1337 cm^{-1}. The two bending vibrations are equivalent and absorb at the same frequency of 667.3 cm^{-1}. The overtone of this is 1334.6 cm^{-1} (2 × 667.3) which is very close to 1337 cm^{-1}. Thus, Fermi Resonance takes place resulting in the shift of first level towards higher frequency. The mutual perturbation of 1337 cm^{-1} (Fundamental) and 1334.6 cm^{-1} (Overtone) gives rise to two bands at 1285.5 cm^{-1} and 1388.3 cm^{-1} having intensity ratio of 1 : 0.9.

Another example of Fermi Resonance is given by aldehydes in which C—H stretching absorption usually appears as a doublet (~2820 cm^{-1} and ~2720 cm^{-1}) due to the interaction between C—H stretching (fundamental) and the overtone of C—H deformation (bending). Fermi Resonance is also shown by the spectrum of *n*-butyl vinyl ether. In this case, the overtone of the fundamental vibration at 810 cm^{-1} chances to coincide with the band at 1640 cm^{-1}. The mixing of the two bands (Fundamental and Overtone) in accordance with Fermi Resonance gives two bands of almost equal intensity at 1640 cm^{-1} and 1630 cm^{-1}.

Similarly, Fermi Resonance is required to explain the doublet in case of certain lactones and cycloketones. In cyclopentanone, the absorption due to carbonyl group occurs at 1746 cm^{-1} and 1750 cm^{-1}. This splitting can be explained by Fermi Resonance. In large molecules, Fermi Resonance is observed only when the normal vibrations and the overtones are in the same area of the molecule and also if certain symmetrical conditions are fulfilled (Two bands should be of the same symmetry).

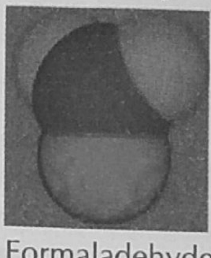
2. Electronic Effects. Changes in the absorption frequencies for a particular group take place when the substituents in the neighbourhood of that particular group are changed. The frequency shifts are due to the electronic effects which include *Inductive effect, Mesomeric effects, Field effects* etc. These effects cannot be isolated from one another and the contribution of one of them can only be estimated approximately. Under the influence of these effects, the force constant or the bond strength changes and its absorption frequency shifts from the normal value. The introduction of alkyl group causes + I effect which results in the lengthening or the weakening of the bond and hence the force constant is lowered and wave number of absorption decreases. Let us compare the wave numbers of ν (C=O) absorptions for the following compounds:

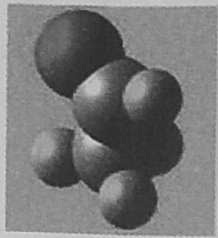
Weaking of bond due to +I effect shift the band towards the ~~higher~~ ~~lower freq, higher~~ lower freq.

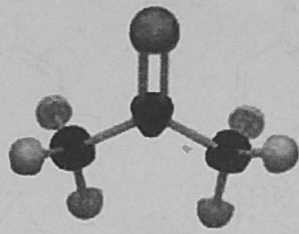
O H-C=O.

(*i*) Formaldehyde (HCHO) 1750 cm^{-1}
(*ii*) Acetaldehyde (CH_3CHO) 1745 cm^{-1}
(*iii*) Acetone (CH_3COCH_3) 1715 cm^{-1}

Ketones < Aldehydes


Formaladehyde


Acetaldehyde


Acetone

Note. Aldehydes absorb at higher wave number than ketones.

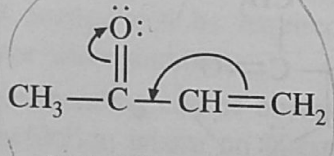
The introduction of an electronegative atom or group causes–I effect which results in the bond order to increase. Thus, the force constant increases and hence the wave number of absorption rises. Now, let us consider the wave numbers of absorptions in the following compounds:

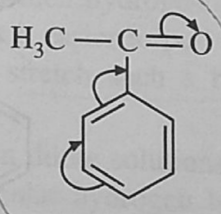
(*i*) Acetone (CH_3COCH_3) 1715 cm^{-1}
(*ii*) Chloroacetone (CH_3COCH_2Cl) 1725 cm^{-1}
(*iii*) Dichloroacetone ($CH_3COCHCl_2$) 1740 cm^{-1}
(*iv*) Tetrachloroacetone ($Cl_2CH—CO—CHCl_2$) 1750, 1778 cm^{-1}

−I effect ↑ then freq ↑

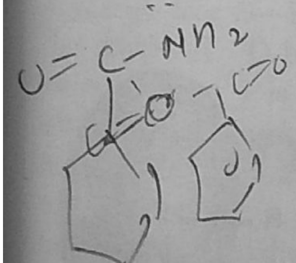
In most of the cases, mesomeric effect works alongwith inductive effect and cannot be ignored. Conjugation lowers the absorption frequency of C=O stretching whether the conjugation is due to α, β-unsaturation or due to an aromatic ring. In some cases, inductive effect dominates over mesomeric effect while reverse holds for other cases. *Mesomeric effect* causes lengthening or the weakening of a bond leading in the lowering of absorption frequency. Consider the following compounds:

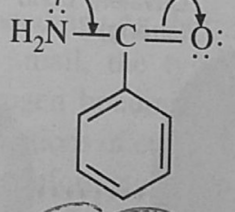
In these two cases, –I effect is dominated by mesomeric effect and thus, the absorption frequency falls.

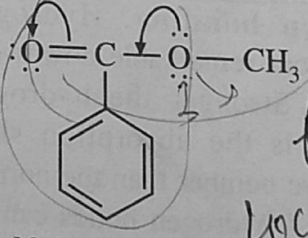

Methyl Vinyl Ketone
ν C = O 1706 cm^{-1}


Acetophenone
ν C = O 1693 cm^{-1}

In some cases, where the lone pair of electrons present on an atom is in conjugation with the double bond of a group, say carbonyl group, the mobility of the lone pair of electrons matters. Let us compare the absorption frequencies of ν(C=O) stretching in amides and esters.



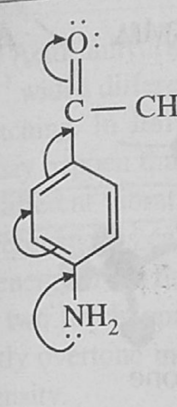

Benzamide
ν C = O 1693 cm^{-1}


Methyl benzoate
ν C = O 1730 cm^{-1}

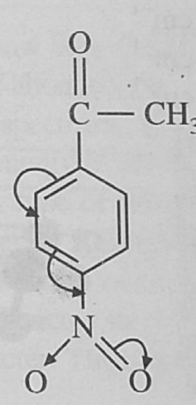
higher conjugation lower freq.

As nitrogen atom is less electronegative than oxygen atom, the electron pair on nitrogen atom in amide is more labile and participates more in conjugation. Due to this greater degree of conjugation, the C=O absorption frequency is much less in amides as compared to that in esters.

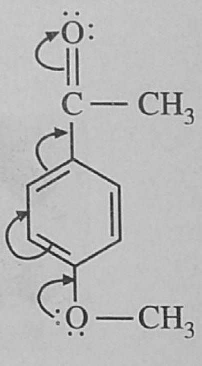
Let us compare the absorption frequencies of the following compounds:



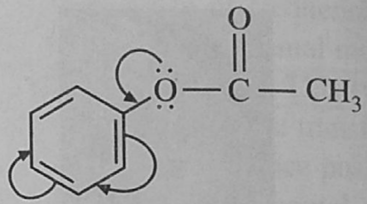
I
p-Amino
Acetophenone
ν C=O 1677 cm^{-1}



II
p-Nitro
Acetophenone
ν C=O 1700 cm^{-1}



III
p-Methoxy
Acetophenone
ν C=O 1684 cm^{-1}

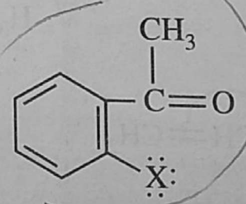


IV
Phenyl Acetate
ν C=O 1770 cm^{-1}

Due to the low electronegativity of nitrogen atom, the lone pair of electrons participates more in conjugation in compound I as compared to that in compound III. Thus, in compound I, ν (C=O) absorption occurs at lower wave number compared to that in compound III. In compounds II and IV, inductive effect dominates over mesomeric effect and hence absorption takes place at comparatively higher frequencies.

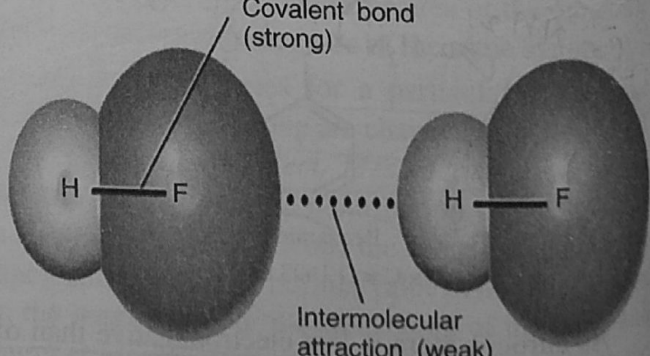
* **It is important to note that only inductive effect is considered when the compound is meta substituted.**

In para substitution, both inductive and mesomeric effects become important and the domination of one over the other will decide the wave number of absorption. In ortho substitution, inductive effect, mesomeric effect alongwith steric effect are considered. In ortho substituted compounds, the lone pairs of electrons on two atoms influence each other through space interactions and change the vibrational frequencies of both the groups. This effect is called **Field effect**. Consider ortho haloacetophenone.



The non-bonding electrons present on oxygen atom and halogen atom cause electrostatic repulsions. This causes a change in the state of hybridisation of C=O group and also makes it to go out of the plane of the double bond. Thus, the conjugation is diminished and absorption occurs at a higher wave number. Thus, for such ortho substituted compounds, cis absorbs (field effect) at a higher frequency as compared to the trans isomer.

3. Hydrogen bonding. Hydrogen bonding brings about remarkable downward frequency shifts. Stronger the hydrogen bonding, greater is the absorption shift towards lower wave number than the normal value. Two types of hydrogen bonds can be readily distinguished in Infra-red technique. Generally, intermolecular hydrogen bonds give rise to broad bands whereas bands arising from intramolecular hydrogen bonds are sharp and well defined. Intermolecular hydrogen



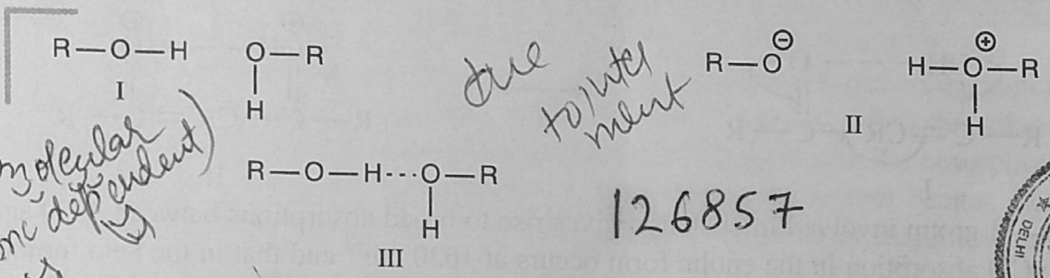
Inter-molecular Hydrogen bonding

Intermolecular broad) Intramolecular Sharp and well defined

bonds are **concentration dependent**. On dilution, the intensities of such bands are independent of concentration. The absorption frequency difference between free and associated molecule is smaller in case of intramolecular hydrogen bonding than that in intermolecular association.

Hydrogen bonding in O—H and N—H compounds deserve special attention. Mostly non-associating solvents like carbon disulphide, chloroform, carbon tetrachloride are used because some solvents like benzene, acetone etc., influence O—H and N—H compounds to a considerable extent. As nitrogen atom is less electronegative than an oxygen atom, hydrogen bonding in amines is weaker than that in alcohols and thus, the frequency shifts in amines are less dramatic. For example, amines show N—H stretching at 3500 cm^{-1} in dilute solutions while in condensed phase spectra, absorption occurs at 3300 cm^{-1}.

In aliphatic alcohols, a sharp band* appears at 3650 cm^{-1} in dilute solutions due to free O—H group while a broad band is noticed at 3350 cm^{-1} due to hydrogen bonded O—H group. Alcohols are strongly hydrogen bonded in condensed phases. These are usually associated as dimers and polymers which result in the broadening of bands at lower absorption frequencies. In vapour state or in inert solvents, molecules exist in free state and absorbs strongly at 3650 cm^{-1}. Alcohols can be written in the following resonating structures.



(Intermolecular conc dependent)

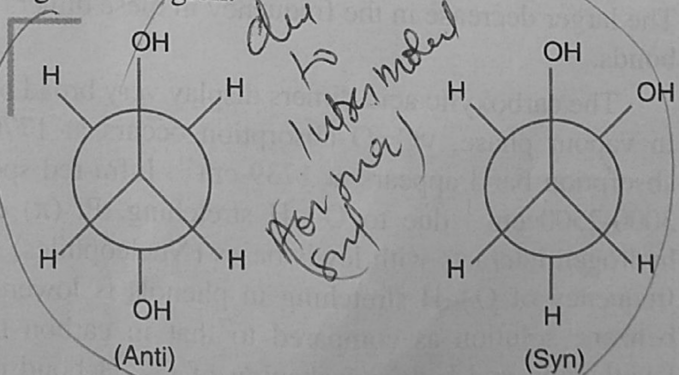
due to inter ment

Fig. 3.9. Resonating structures of alcohols.

Structure III is the hybrid of structures I and II. This results in the lengthening of the original O—H group. The electrostatic force of attraction with which hydrogen atom of one molecule is attracted by the oxygen atom of another molecule makes it easier to pull hydrogen away from the oxygen atom. Thus, small energy will be required to stretch such a bond (O—H) and hence absorption occurs at a lower wave number.

Intramolecular hydrogen bonding can be observed in dilute solutions of di- and polyhydroxy compounds in carbon tetrachloride where no intermolecular hydrogen bonds are formed. Under these conditions, it was observed that a number of cyclic and acyclic diols have two bands and others have a single band in the O—H stretching mode region.

The spectrum of glycol in dilute carbon tetrachloride shows two νO—H bands at 3644 cm^{-1} and 3612 cm^{-1}. The band at 3644 cm^{-1} is due to free O—H and that at 3612 cm^{-1} is due to O—H···O bonding. As the absorption shift (32 cm^{-1}) is small, the type must be intramolecular hydrogen bonding. Out of the two possible conformations of ethylene glycols, only staggered syn conformation can form an intramolecular hydrogen bond. The occurrence of a strong band arising from the intramolecular



due to Intermolecular (H-bonding)

Fig. 3.10. Anti and Syn conformations of Ethylene glycol.

* The band due to free O—H group is usually less intense while that due to bonded O—H group is broad and sharp.

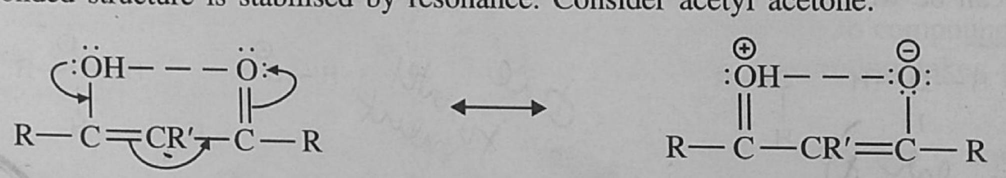
hydrogen bond shows that the molecule exists in staggered-syn conformation.

The energy of the hydrogen bond compensates for excess energy (5 kcal mole^{-1}) due to steric and dipolar repulsions of the two O—H groups. The small frequency shift ($\Delta\nu$ = 32 cm^{-1}) shows that hydrogen bonding is weak.

At higher concentrations, the I.R spectra of diols show a third band arising from the intermolecular hydrogen bonding of the hydroxy group. This new band shows all the characteristics observed in monohydric alcohols, *i.e.*, the bands formed are strong and broad. The intermolecular association of diols results in a larger shift of νO—H than intramolecular hydrogen bonding does. If the free νO—H in diols is 3626 cm^{-1}, then intermolecularly bonded O—H abosrbs at 3477 cm^{-1}.

Note. In aromatic amines, differences between intramolecularly associated and free N—H bond absorption frequencies are small and difficult to observe.

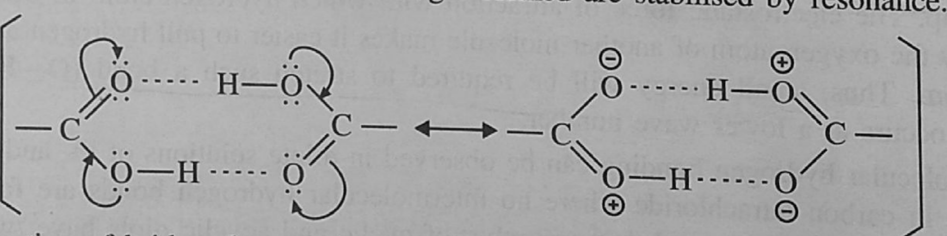
In enols and chelates, hydrogen bonding is exceptionally strong and absorptions due to O—H stretching occurs at very low values. As these bonds are not broken easily on dilution by an inert solvent, free O—H stretching may not be seen at low concentrations. It is due to the fact that the bonded structure is stabilised by resonance. Consider acetyl acetone:



I II

The O—H group involved in chelation gives rise to broad absorptions between 3000 and 2500 cm^{-1}. The νCO absorption in the enolic form occurs at 1630 cm^{-1} and that in the keto form at 1725 cm^{-1}. From the intensities of the two peaks, it is possible to determine the quantities of the enolic and the ketonic forms.

Mostly the acids exist as dimers and bridges formed are stabilised by resonance.



The formation of bridges lowers the force constants and thus νC=O and νO—H decreases. The larger decrease in the frequency in these dimers indicates the exceptional strengths of hydrogen bonds.

The carboxylic acid dimers display very broad band at 3000-2500 cm^{-1} due to O—H stretching. In vapour phase, νC=O absorption occurs at 1770 cm^{-1} in acetic acid and in the liquid state, absorption band appears at 1739 cm^{-1}. Infra-red spectrum of benzoic acid shows a broad band at 3000-2500 cm^{-1} due to O—H stretching. Pi (π) cloud interactions are also noted when acidic hydrogen interacts with lewis bases (Nucleophiles) such as alkenes and benzene. For example, the frequency of O—H stretching in phenols is lowered by 40-100 cm^{-1} when spectrum is taken in benzene solution as compared to that in carbon tetrachloride solution. Due to this interaction, lengthening and hence weakening of O—H bond occurs.

4. Bond angles. It has been found that the highest νC=O frequencies arise in the strained cyclobutanones. This can be explained in terms of bond angles strain. The C—CO—C bond angle is reduced below the normal angle of 120° and this leads to increased *s*-character in the C=O bond. Greater *s*-character causes shortening of C=O bond and thus C=O str occurs at higher frequency.

In case the bond angle is pushed outwards above 120°, the opposite effect operates. Due to this reason, di-tert-butyl ketone absorbs at 1697 cm^{-1} (low) as a result of C=O str.

It is also assumed that there is no change in the C=O force constant and it is an increased rigidity in the C—CO—C bond system as the ring size decreases. In such cases, C=O str couple more effectively with C—C str leading to higher C=O str frequencies.

to produce infrared Red Radial

3.8 Scanning of Infra-red Spectrum (Instrumentation)

The most important source of Infra-red light for scanning the spectrum of an organic compound is **Nernst glower** which consists of a rod of the sintered mixture of the oxides of Zirconium, Ytterium and Erbium. The rod is electrically heated to 1500°C to produce Infra-red radiations.

A rod of silicon carbide (Globar) can also be electrically heated to produce Infra-red radiations. To obtain monochromatic light, optical prisms or gratings can be used. For prism material, glass or quartz cannot be used since they absorb strongly through most of the Infra-red region. Sodium chloride or certain alkali metal halides are commonly used as cell containers or for prism materials as these are transparent to most of the Infra-red region under consideration. Sodium chloride is hygroscopic and is, therefore, protected from the condensation of moisture by working at suitable temperature. Gratings give better resolution than do prisms at high temperatures.

Light from the source is split into two beams. One of the beams is passed through the sample under examination and is called the sample beam. The other beam is called the reference beam. When the beam passes through the sample, it becomes less intense due to the absorption of certain frequencies. Now, there will be a difference in the intensities of the two beams. Let I_0 be the intensity of the reference beam and I be the intensity of the beam after interaction with the sample respectively.

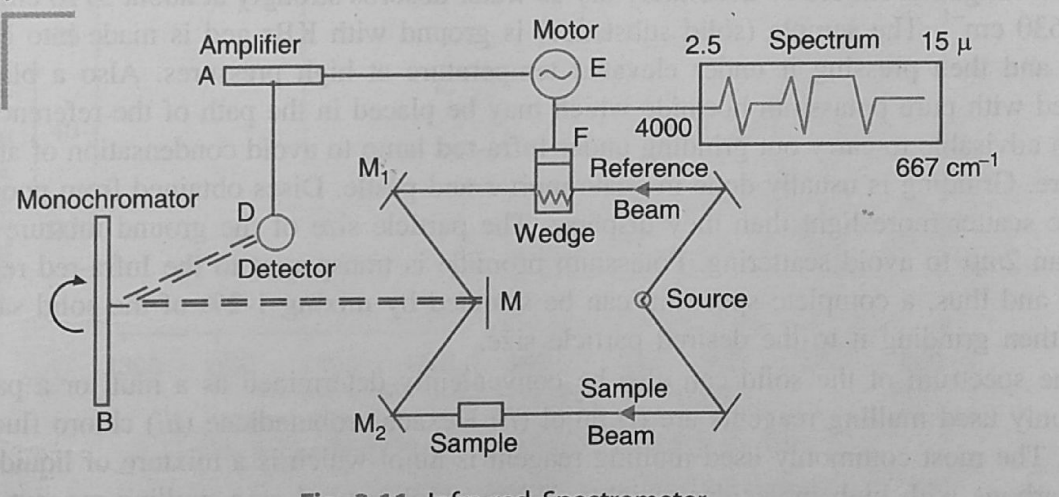


Fig. 3.11. Infra-red Spectrometer.

The absorbance of the sample at a particular frequency can be calculated as:

$$A = \log (I_0/I)$$

Also transmittance, $$T = I/I_0$$

or $$A = \log (1/T)$$

Intensities of the bands can be recorded as a linear function T against the corresponding wave-number. Intensities of the two beams are converted into and measured as electrical energies with the help of detector thermopile. For this we proceed as follows:

* It works on the thermopile principle—when two ends of the metal wire are attached with the two ends of another metal wire, then a difference in temperature between the two ends causes a current to flow in the wires. The intensity of radiation falling on the thermopile produces a current.

The two beams (Sample and Reference) are made to fall on a segmented mirror M (chopper) with the help of two mirrors M_1 and M_2. The chopper (M) which rotates at a definite speed reflects the sample and the reference beams to a monochromator grating (B). As the grating rotates slowly, it sends individual frequencies to detector thermopile* (D) which converts Infra-red energy into electrical energy. It is then amplified with the help of amplifier (A). Due to the difference in intensities of the two beams, alternating currents start flowing from the detector thermopile to the amplifier. The amplifier is coupled to a small motor (E) which drives an optical wedge (F). The movement of the wedge is in turn coupled to a pen recorder which draws absorption bands on the calibrated chart. The movement of the wedge continues till the detector receives light of equal intensity from the sample and the reference beams. Calibration can be carried out using spectrum of Polystyrene.

3.9 Sampling Techniques

8

Various techniques can be employed for placing the sample in the path of the Infra-red beam depending upon whether the sample is a gas, a liquid or a solid. The intermolecular forces of attraction are most operative in solids and least in case of gases. Thus, the sample of the same substance shows shifts in the frequencies of absorption as we pass from the solid to the gaseous state. In some cases, additional bands are also observed with the change in the state of the sample. Hence, it is always important to mention the state of the sample on the spectrum which is scanned for its correct interpretation.

(*a*) Solids. Solids for the Infra-red spectrum may be examined as an alkali halide mixture. Alkali metal halides, usually sodium chloride, which is transparent throughout the Infra-red region is commonly used for the purpose. Potassium bromide also serves the purpose well. The substance under investigation should be absolutely dry as water absorbs strongly at about 3710 cm^{-1} and also near 1630 cm^{-1}. The sample (solid substance) is ground with KBr and is made into a disc after drying and then pressing it under elevated temperature at high pressures. Also a blank disc is prepared with pure potassium bormide which may be placed in the path of the reference beam. It is often advisable to carry out grinding under Infra-red lamp to avoid condensation of atmospheric moisture. Grinding is usually done in agate mortar and pestle. Discs obtained from poorly ground mixture scatter more light than they disperse. The particle size of the ground mixture should be less than 2$m\mu$ to avoid scattering. Potassium bromide is transparent to the Infra-red region (2.5μ – 15μ) and thus, a complete spectrum can be scanned by mixing 1-2% of the solid sample with it and then grinding it to the desired particle size.

The spectrum of the solid can also be conveniently determined as a mull or a paste. Some commonly used mulling reagents are (*i*) nujol (*ii*) Hexachlorobutadiene (*iii*) chloro fluoro carbon oil etc. The most commonly used mulling reagent is nujol which is a mixture of liquid paraffinic hydrocarbons with high molecular weights. When nujol is used as a mulling reagent [Spectrum shown in figures 3.12 (*a*) and 3.12 (*b*)] absorptions due to C—H stretching at 3030-2860 cm^{-1} and those for C—H bending at about 1460 cm^{-1} and 1374 cm^{-1} are observed. Clearly, no information about the sample can be recorded from these regions of absorptions. The regions of absorptions due to nujol can be studied only by taking another spectrum of the sample using hexachlorobutadiene as the mulling reagent. A solid film can also be deposited over the alkali metal halide (NaCl or KBr) disc by putting a drop of the solution of the sample (sample dissolved in a volatile solvent) on the disc and then evaporating the solvent. Polymers, fats and waxy materials show excellent spectra in this way. Solid samples can also be examined in solutions. A polystyrene film is commonly used for calibration of wave numbers. For this, calibration is done using the bands at 3026, 3002, 2924, 1602, 1495 and 906 cm^{-1}.

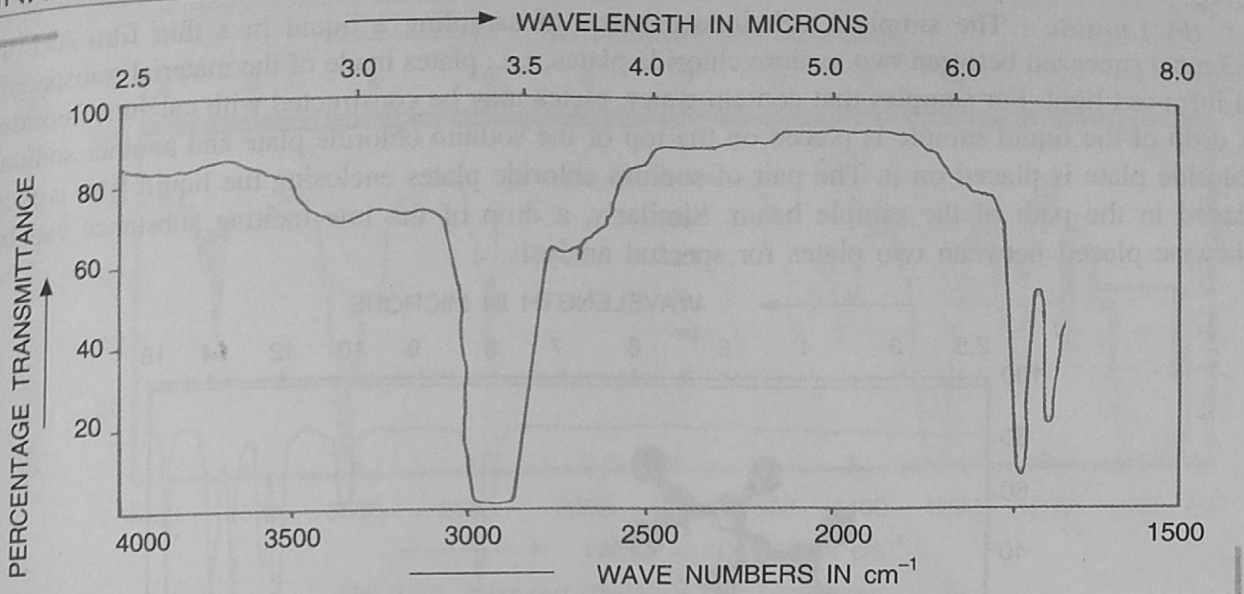


Fig. 3.12. (*a*) Infra-red spectrum of Nujole (higher).

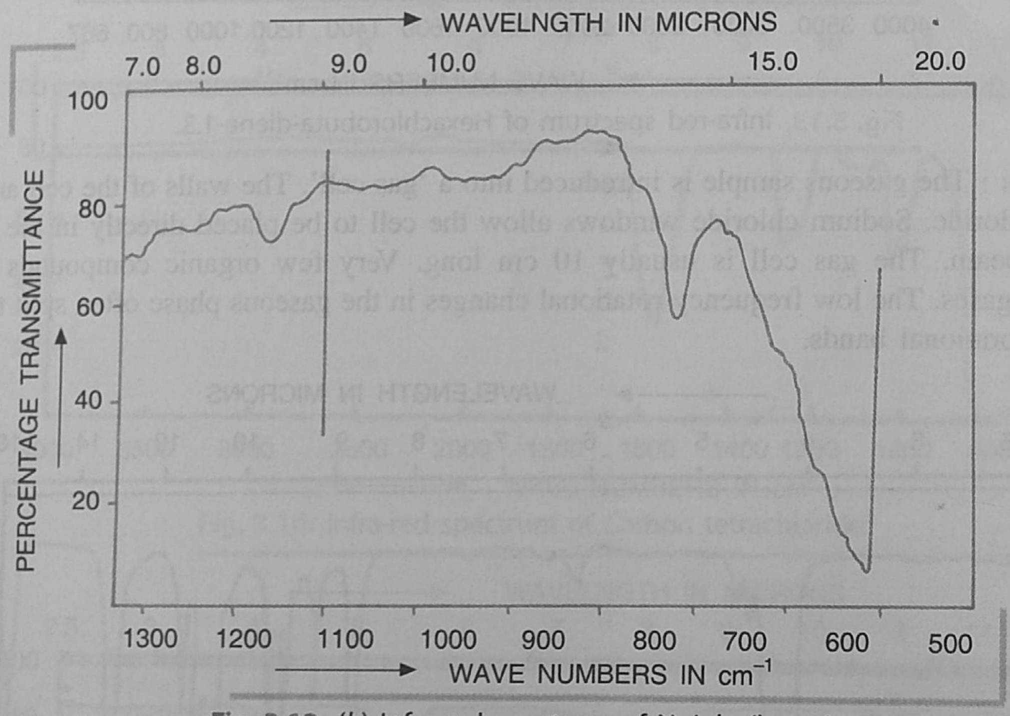


Fig. 3.12. (*b*) Infra-red spectrum of Nujole (lower).

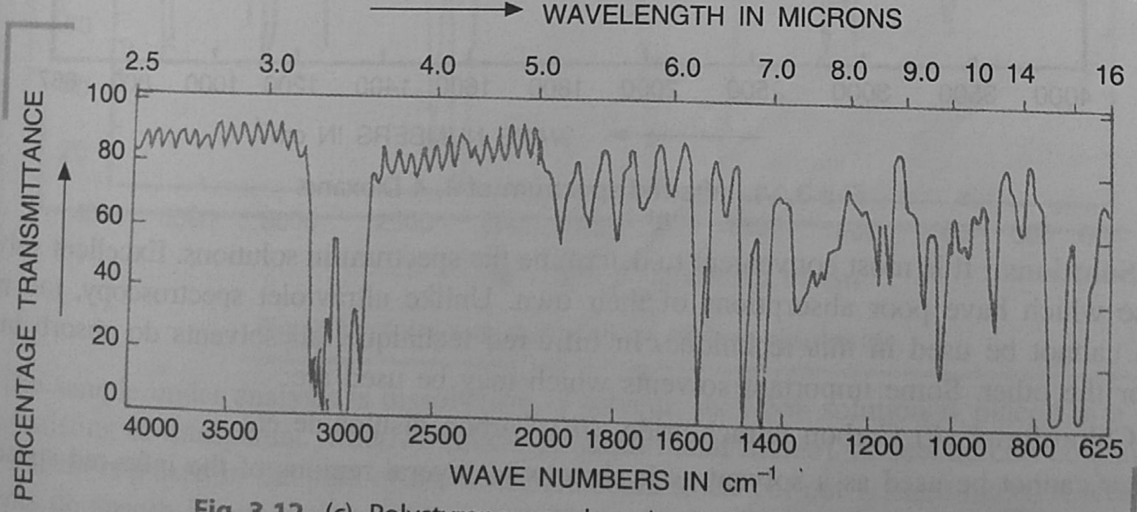


Fig. 3.12. (*c*). Polystyrene-wavelength v/s Percent transmittance.

(*b*) **Liquids :** The simplest technique consists of sampling a liquid in a thin film (0.1 to 0.3 mm) squeezed between two sodium chloride plates, *i.e.*, plates made of the material transparent to Infra-red light. For samples that contain water, plates may be constructed with calcium fluoride. A drop of the liquid sample is placed on the top of the sodium chloride plate and another sodium chloride plate is placed on it. The pair of sodium chloride plates enclosing the liquid film is then placed in the path of the sample beam. Similarly, a drop of the low melting substance can be likewise placed between two plates for spectral analysis.

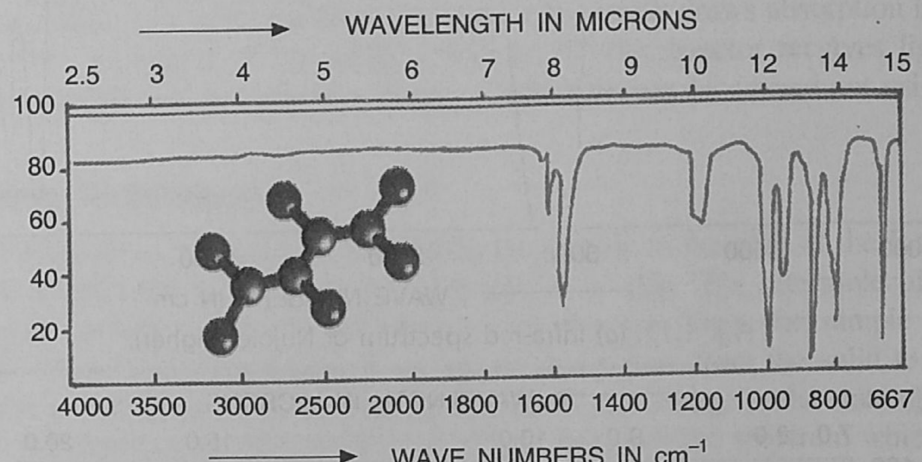


Fig. 3.13. Infra-red spectrum of Hexachlorobuta-diene-1.3.

(*c*) **Gases :** The gaseous sample is introduced into a 'gas cell'. The walls of the cell are made of sodium chloride. Sodium chloride windows allow the cell to be placed directly in the path of the sample beam. The gas cell is usually 10 cm long. Very few organic compounds can be examined as gases. The low frequency rotational changes in the gaseous phase often split the high frequency vibrational bands.

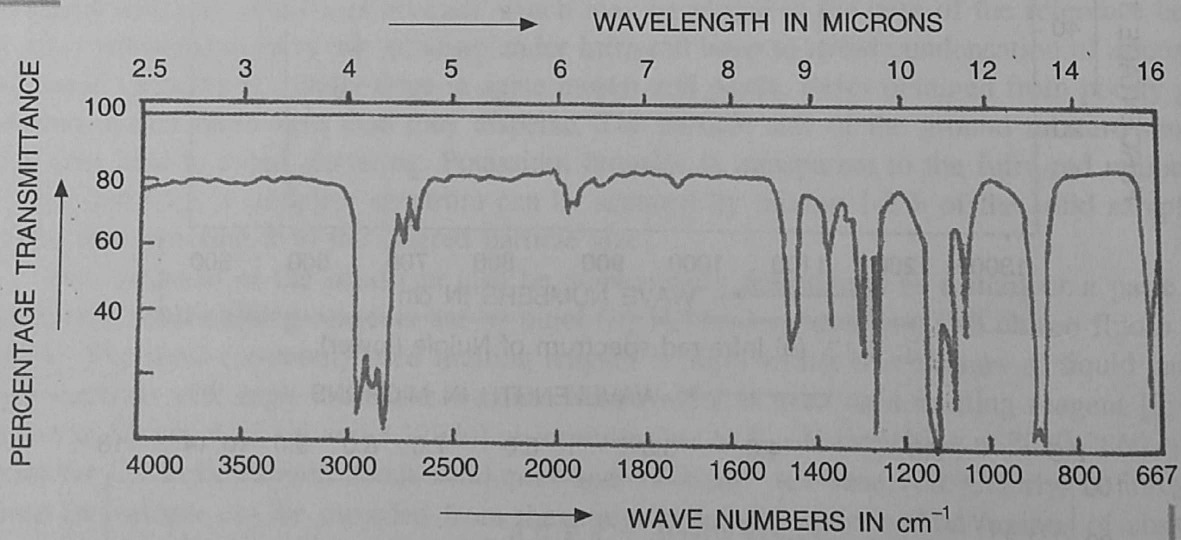


Fig. 3.14. Infra-red spectrum of 1, 4 Dioxane.

(*d*) **Solutions :** It is most convenient to determine the spectrum in solutions. Excellent solvents are those which have poor absorptions of their own. Unlike ultraviolet spectroscopy, too many solvents cannot be used in this technique. In Infra-red technique, all solvents do absorb in one region or the other. Some important solvents which may be used are:

(*i*) Chloroform, (*ii*) Carbon tetrachloride, (*iii*) Carbon disulphide etc.

Water cannot be used as a solvent as it absorbs in several regions of the infra-red spectrum.

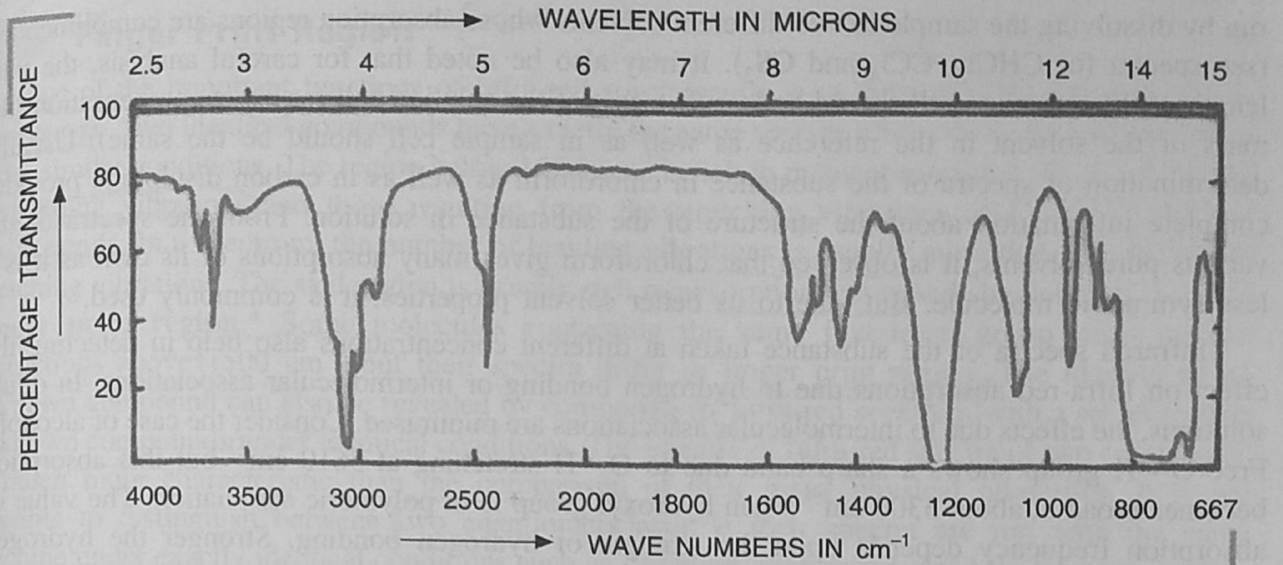


Fig. 3.15. Infra-red spectrum of Chloroform.

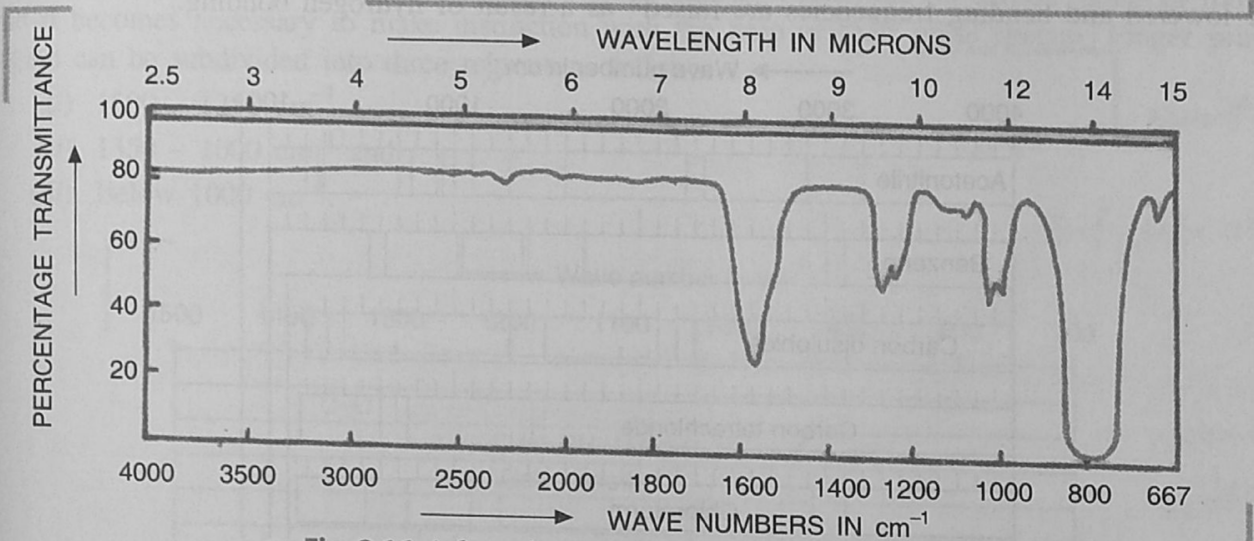


Fig. 3.16. Infra-red spectrum of Carbon tetrachloride.

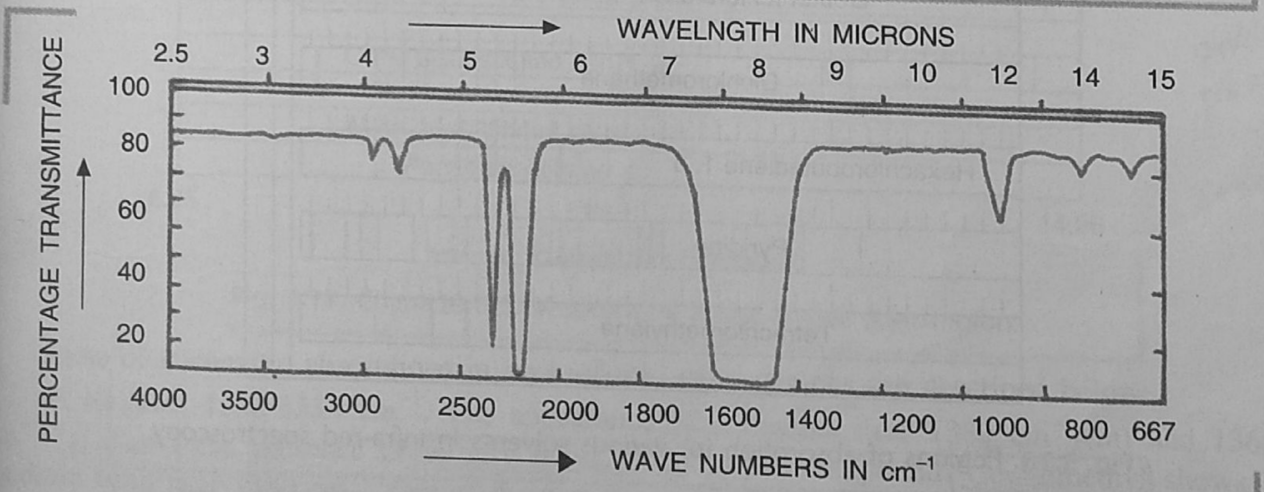


Fig. 3.17. Infra-red spectrum of Carbon disulphide.

The sample under analysis is dissolved in a solvent. Its 1-5% solution is placed in a solution cell consisting of transparent windows (made of alkali metal halide). A second cell containing the pure solvent is placed in the path of the reference beam to cancel out solvent interferences. In fact solvents do absorb in one region or the other. Hence, for correct analysis, two spectra should be run by dissolving the sample in two different solvents whose absorption regions are complimentary (see spectra for $CHCl_3$, CCl_4 and CS_2). It may also be noted that for careful analysis, the path length of the reference cell should be less than that of the solution cell for exact compensation (the mass of the solvent in the reference as well as in sample cell should be the same). Usually, determination of spectra of the substance in chloroform as well as in carbon disulphide provides complete information about the structure of the substance in solution. From the spectra of the various pure solvents, it is observed that chloroform gives many absorptions of its own as it is a less symmetric molecule. But, due to its better solvent properties, it is commonly used.

Infrared spectra of the substance taken at different concentrations also help in detecting the effect on Infra-red absorptions due to hydrogen bonding or intermolecular associations. In dilute solutions, the effects due to intermolecular associations are minimised. Consider the case of alcohols. Free O—H group shows a sharp band due to O—H stretching at 3610 cm^{-1} but this absorption becomes broad at about 3300 cm^{-1} when hydroxyl group is in polymeric association. The value of absorption frequency depends upon the strength of hydrogen bonding. Stronger the hydrogen bonding, lower is the stretching frequency of absorption. It may be noted that stretching frequencies are lowered and bending frequencies are raised* as a result of hydrogen bonding.

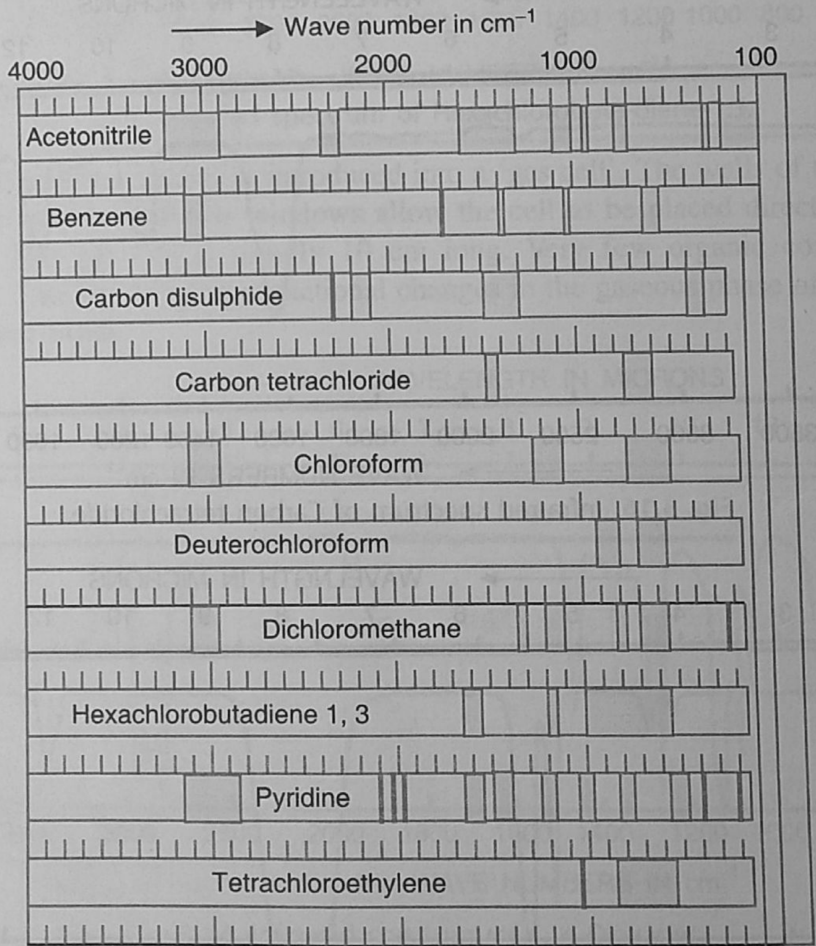


Fig. 3.18. Regions of absorption for various solvents in infra-red spectroscopy.

* It involves the change in coplanarity which is comparatively difficult to achieve and requires greater energy.

* N—H bending vibrations(an exception) occurs above 1600 cm^{-1}. For example, N—H bending in $—NH_2$ group occurs between 1650-1500 cm^{-1} (*m*).

... absorption is called finger print

3.10 Finger Print Region

One of the important functions of Infra-red spectroscopy is to determine the identity of two compounds. Two identical compounds have exactly the same spectra when run in the same medium under similar conditions. The region below 1500 cm^{-1} is rich in many absorptions which are caused by bending vibrations and those resulting from the stretching vibrations of C—C, C—O and C—N bonds. In a spectrum, the number of bending vibrations is usually more than the number of stretching vibrations. The said region is usually rich in absorption bands and shoulders. It is called **Finger print region**.* Some molecules containing the same functional group show similar absorptions above 1500 cm^{-1} but their spectra differ in finger print region. The identity of an unknown compound can also be revealed by comparing its Infra-red spectrum with a set of spectra of known compounds under identical conditions. The identity of Infra-red spectra of two compounds is much more characteristic than the comparison of their many physical properties. It is not possible to distinguish between two enantiomers even if their spectra are run with the same machine under exactly identical conditions such as sampling, scan speed etc. Also it is not possible to distinguish the Infra-red spectra of straight chain alkanes containing 30 carbon atoms or more and it becomes necessary to make distinction with the help of their mass spectra. Finger print region can be subdivided into three regions as follows:

(*i*) 1500 – 1350 cm^{-1}

(*ii*) 1350 – 1000 cm^{-1} and

(*iii*) Below 1000 cm^{-1}.

large no. Absorption vari freq

Diff type of IR Absorp

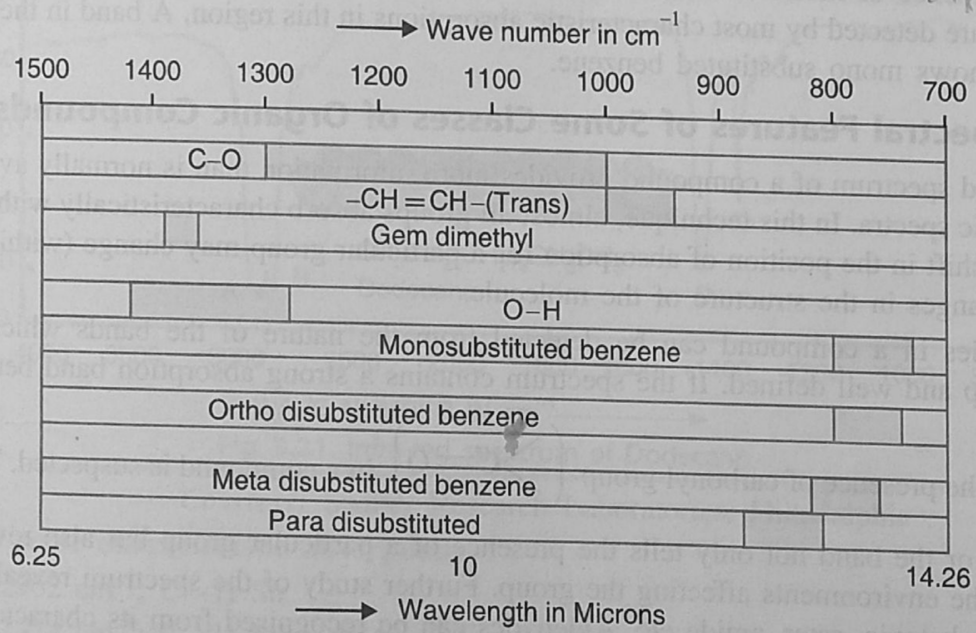


Fig. 3.19. Characteristic absorptions in the Finger print region.

1500–600 cm^{-1} Skeltan even isomer none same absorption band print

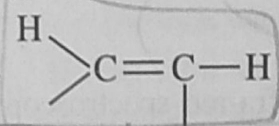
polarity

Some characteristic absorptions in each of the above regions are described below:

(*i*) **Region: 1500-1350 cm^{-1}**. The appearance of a doublet near 1380 cm^{-1} (m) and 1365 cm^{-1} (s) shows the presence of tertiary butyl group in the compound. Gemdimethyl shows a medium band near 1380 cm^{-1}. Out of the two strong bands for the nitro compound, one appears in the finger print region, near 1350 cm^{-1}.

(*ii*) **Region: 1350-1000 cm^{-1}**. All classes of compounds viz. alcohols, esters, lactones, acid anhydrides show characteristic absorptions (strong bands) in this region due to C—O stretching. Primary alcohols form two strong bands at 1350-1260 cm^{-1} and near 1050 cm^{-1}. Phenols absorb near 1200 cm^{-1}. Esters show two strong bands between 1380-1050 cm^{-1}. Absorption in the region 1150-1070 cm^{-1} is most characteristic of ethers, *i.e.*, C—O stretching in C—O—C group. 01-1200

ether

(*iii*) Region: below 1000 cm^{-1}.  deformation at ~700 cm^{-1} (s) and that at 970-960 cm^{-1} (s) distinguishes between cis and trans alkenes. The higher value indicates that the hydrogen atoms in the alkene are trans with respect to each other.

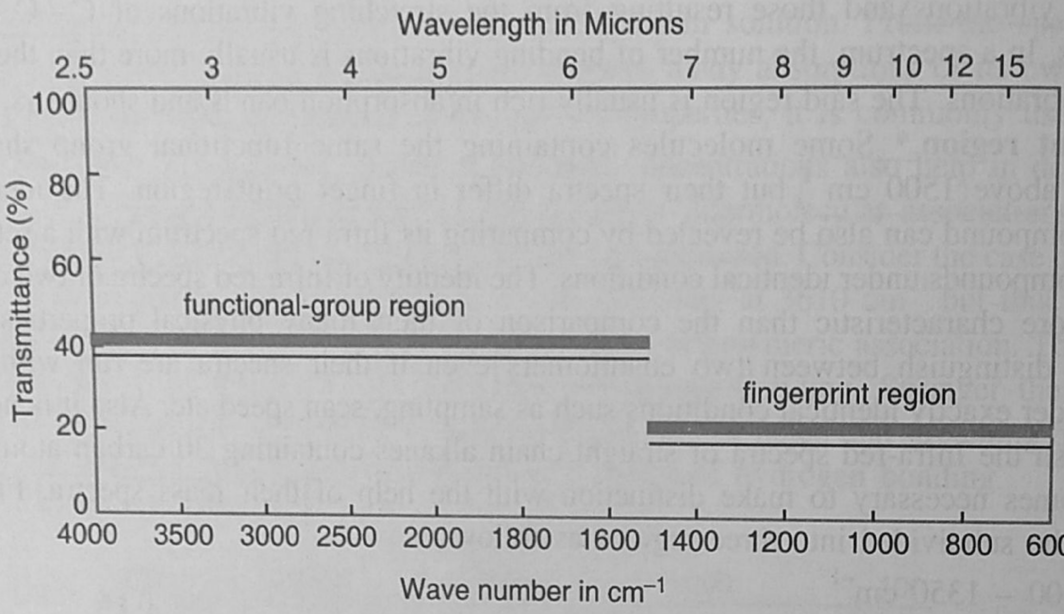


Fig. 3.20. Functional group and finger print regions.

The presence of mono substitution and also disubstitution at ortho, meta and para positions in benzene are detected by most characteristic absorptions in this region. A band in the region 750-700 cm^{-1} shows mono substituted benzene.